INCH-POUND

MIL-C-24643/43C
14 March 1994
SUPERSEDING
MIL-C-24643/43B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS TYPES, LS2SJ, LS3SJ AND LS4SJ (INCLUDING VARIATION LS2SJA, LS3SJA AND LS4SJA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LSSJ (unarmored) and LSSJA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First	- Copper conductor, tin coated, ASTM B 286, sizes 14 and smaller. Uncoated for AWG sizes 12, class B stranding and larger.
Second	- Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	- Ethylene propylene rubber or cross-linked polyethylene insulation. (see table I for dimensions). Standard identification code applied by method 3.
Fourth	- Two, three or four conductors, as required, cabled together with fillers to form a firm, well rounded assembly.
* Fifth	- Braided shield of tin coated copper. Minimum coverage of 85 percent.
Sixth	- Binder tape of polyester, type G in accordance with MIL-I-631, applied helically with overlap.

Construction variant for type LSSJ

* Seventh	- Cross-linked polyolefin jacket. (Minimum average wall thickness 0.035 inch.)

AMSC N/A FSC 6145

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Construction variant for type LSSJA

* Seventh - Cross-linked polyolefin jacket. (Minimum, average wall thickness 0.035 inch).

* Eighth - Braided metal armor

TABLE I. Details.

Military part no. M24643/43 1/	Conductors size				Insulation wall thickness (min avg) (inch)	Lay of conductor maximum (inches)	Type's overall diameter (inches)				Conductor resis
	Size 2/	No.	Size 3/				LSSJ		LSSJA		
			ASTM B 286	ASTM B 8			min	max	min	max	
-01UU	22	2	22-19		0.016	1-1/2	0.261	0.275	0.311	0.325	16.54
-02UU	20	2	20-19		.016	1-1/2	.273	.290	.323	.340	10.16
-03UU	18	2	18-19		.016	1-1/2	.295	.310	.345	.360	6.47
-04UU	16	2	16-19		.016	2-1/2	.309	.325	.359	.375	5.02
-05UU	14	2	14-19		.016	2-1/2	.337	.350	.387	.400	3.17
-06UU	12	2		12(class B)	.024	3	.417	.430	.467	.480	1.72
-07UU	11	2		10(class B)	.025	3	.447	.460	.495	.510	1.085
-08UU	9	2		9(class B)	.031	4	.525	.545	.575	.595	.86
-09UU	7	2		7(class B)	.032	4	.600	.615	.650	.665	.54
-10UU	22	3	22-19		.016	1-1/2	.271	.285	.321	.335	16.54
-11UU	20	3	20-19		.016	1-1/2	.284	.300	.334	.350	10.16
-12UU	18	3	28-19		.016	1-1/2	.308	.325	.358	.375	6.47
-13UU	16	3	16-19		.016	2-1/2	.323	.340	.373	.390	5.02
-14UU	14	3	14-19		.016	2-1/2	.353	.370	.303	.320	3.17
-15UU	12	3		12(class B)	.024	3	.440	.455	.490	.505	1.72
-16UU	9	3		9(class B)	.031	4	.594	.620	.644	.670	.86
-17UU	20	4	20-19		.032	1-1/2	.303	.320	.353	.370	10.16
-18UU	16	4	16-19		.016	2-1/2	.346	.360	.396	.410	5.02
-19UU	14	4	14-19		.016	3	.380	.395	.430	.445	3.17

1/ Unarmored, overall shield. When armored option is required, see MIL-C-24643 for configuration of part number.

2/ Type LSSJ and LSSJA.

3/ Size designation does not correlate to conductor size for all types and sizes.

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - omhs/1000 feet at 25°C, maximum.....	(see table I)
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	1200
Insulation resistance - Megohms/1000 feet minimum.	
Conductor to shield	500
Jacket flaws	No failure
Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Group B:	
* Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Ehylene propylene	
Air Oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged minimum	75
Cross-linked polyethylene	
Air Oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80

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	Jacket (cable)	
	Air oven	
	Tensile strength - percent of unaged, minimum	60
	Elongation - percent of unaged minimum	60
	Hot oil immersion	
*	Tensile strength - percent of unaged, minimum	50
*	Elongation - percent of unaged, minimum	50
	Shrinkage	No failure
	Heat distortion - percent of unaged, maximum	39
	*Permanence of printing (jacket) - (type LSSJ only) cycles, minimum	125
	Armor (type LSSJA only) - conformance to material, construction and coverage	No failure
	Shield - conformance to material, construction and coverage	No failure
	Surface transfer impedance (type LSMUS only)	
	* Milliohms per meter, maximum	700
	Emp response time - db, minimum	60

Group D:

Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

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UNIT ORDERING LENGTHS: 1000 feet nominal.

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Navy - SH

Agent:
DLA-IS

(Project No. 6145-2057-43)

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